Building Urban Playgrounds for Video Games

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Introduction

My name is Lincoln Hughes, and I've been an environment artist in the industry for roughly 11 years. I'm currently at Warner Bros. Games Montreal, but in the past I've worked at Electronic Arts, Ubisoft, Relic, Next Level and Piranha. The most notable games I've worked on include Watch Dogs, Watch Dogs 2, Far Cry 4, Star Wars: The Old Republic, Luigi's Mansion: Dark Moon, Ghost Recon: Future Soldier and SOCOM: Confrontation.

Starting at the age of four, I drew Ninja Turtles and eventually found a love for all things dinosaurs after the release of Jurassic Park. Put simply, my life changed. I remember being absolutely obsessed with dinosaurs—so much so that I would sketch them every night before falling asleep, slowly improving my drawing skills!

My older brother brought me in one day to see the game studio he worked at, and I remember being in awe. "People get paid to do this?" I thought. The owner of his game studio gave me a copy of Maya at the age of 12 and I got to work learning 3D.

Six years later, I went to the Art Institute of Burnaby and took their Game Art and Design program.

Mapping out a city

In a perfect world, this is how the blockout process would go:

Research everything. Talk to the other artists, designers and writers working on your game. Figure out what the story is and begin planning how you're going to introduce that lore into the world. Research anything that you can about why the experience will be fun as well as understand the core fantasy of the game. Study endless reference pictures, drilling the art style into your brain. Once you've done all that, you're ready to come up with a plan for your city.

Plan out your city with a 2D map. Learn where the gameplay locations take place and plan out your roads and shortcuts accordingly. Research previously validated references and plan out what types of districts are in the city, even considering tiny details like types of stores and businesses for each street. Having a proper plan in place before the implementation of any 3D assets will help with asset planning and time estimates for your city. Coordinate with your tech team to determine the best way of implementing certain assets into the world (props on roads, vegetation, etc.) and adapt your plan to that method. Validate everything repeatedly with all relevant departments.

Blockout your environments with minimal detail and proper metrics. Build them with the expectation that they will change. Remaining adaptable is incredibly important—as new tools get finished, proper budgets come in and gameplay gets fully discovered and integrated—you'll be changing your art over and over. Keeping your art simple, using literal (metric-friendly) boxes at the beginning is key, but just building your environments modularly, with proper metrics for gameplay, is a must. Quickly flesh out your composition so that everything funnels the player toward the most important gameplay locations. Test everything repeatedly with your designer and tweak accordingly. And once again, validate!

Detail slowly. Don't rush into anything unless you're doing a quick benchmark. Stick to big props that have some form of gameplay attached to them or affect the composition in a grand way. Validate with design and art repeatedly, and test it until your eyes bleed.

Finalize your city with the pretty stuff, adding more and more detail, layer by layer. This is usually done in the last few months of production, as everything has the potential to change repeatedly over the course of development. And when I say everything, I mean everything. Gameplay, tools, art direction, story and more. Be adaptable!

How do gameplay mechanics influence the way you work on the playground?

Having the art in your level completely mesh with gameplay is hands down the most difficult yet important aspect of the level artist's job. Making it pretty is great but nobody will care if it doesn't play well. Here are a few tips to assist in that process.

Metrics play a huge part in the design of an open world. Similar to the way that an architect will need to build things with specific rules in mind, we as artists need to approach the design of our city in the same way. For example, roads and alleyways need to maintain a specific width so they'll be fun to drive along while areas of cover need to be wide and tall enough for players to hide behind. If we violate these rules and build our cities without them, gameplay won't function properly and our levels won't be enjoyable to play.

If there's potential for a gunfight to systematically break out anywhere in the open world, then we need to make almost every square inch of it capable of such a gameplay experience. Normal props—garbage cans, benches, railings and even hedges—become design objects that can be used to benefit the player in some way during gameplay.

You might need to take cover behind it, drive through it, hack it, shoot it or any other number of actions so it can become quite complicated to lay these areas out in a way that works for everybody. My advice is to keep it simple and work with your designers to iron out the kinks. They'll know what it needs to play better much in the same way that you'll know what it needs to look nicer. In an ideal world, artists would know more about design, and designers would know more about art. But in our world, you'll have to compromise and learn from each other at the end of the day.

Choice should always play a significant role in the design of your environments. Some players like to smash their way through the world with guns blazing. Others like to take their time and use stealth to remain undetected. Keeping all of the different playstyles in mind while you compose your environments is incredibly important. Work with your level designer to figure out what choices the player will need to have, and then brainstorm how those choices can be artistically integrated into the reality of the game world.

Modularity. Keep your assets on a grid and make them out of pieces that can be shuffled and rearranged to create new environments quickly and easily. I can't stress this enough—gameplay will evolve, and your environments will become outdated multiple times over the course of production. Expect it and work accordingly.

Why are landmarks important to have in city landscapes?

Landmarks are extremely important to a game world for a number of reasons. For one, they help reduce player confusion by giving gamers visual cues for navigation as well as defining a sense of scale. For example, most people know roughly how big the Eiffel Tower is in Paris. Including it within a game world gives players both a sense of location and a sense of scale.

They can also be used for the integration of story and lore or serve as a distant goal that players slowly work toward over the course of a game. Many times, main missions occur within or around prominent landmarks so it's important to choose and create them wisely. I've played tons of open-world games where I've travelled to some spire in the distance for hours only to eventually (and sadly!) realize upon my arrival that there was nothing to do once I got there. If travelling there was fun, then I might keep playing; but, if not, the game is usually done for me.

Strong landmarks can be a great way to lead the player throughout a world and encourage exploration. If it's possible to travel to those locations—players will—so make sure there's some meaningful payoff waiting their arrival.

How did you work on the skyline?

More important than the look of a skyline is the placement of unique gameplay locations within it. Spacing things out so that areas of gameplay are placed appropriately around the city is key, and the composition of the skyline will usually be dictated by that.

Just like the way a painter will use leading lines, colors, silhouette, lighting and contrast to draw the viewer's eyes to a focal point, we must do the same thing as level artists. Make gameplay locations your focal points: arrange roads so they point at them, sculpt mountains to lead to them and make colors more discordant around them. By doing this, you'll make the most interesting areas of your game more enticing to visit.

From a more visual standpoint, if you're building a city with an instantly recognizable skyline, like New York, it's important to integrate as many iconic, visual cues as possible, even if the in-game city is a condensed version of the real-world city.

How should architecture look and feel in open world games? How do you make buildings real and important in an open world?

I think the answer to those questions depends largely on what type of city you're trying to build. Is it a city that's based in reality, like Chicago? Is it entirely fantastical? Every game is different, and the approach that you take for it varies widely depending on the game's art and gameplay direction. For instance, in a game like Assassin's Creed where the player is constantly climbing walls, buildings and other types of architecture need to be specifically designed for those interactions.

With Watch Dogs 2 taking place in San Francisco, Google Earth was indispensable. Our art staff used it almost every day to plan out entire streets, including the shapes and sizes of buildings, their architecture and the assets they were populated with. With memory budgets in mind, the most frequently occurring architectural styles in San Francisco were chosen and kits for several unique building types were created. After tech artists integrated those pieces into the editor, artists chose which style fit their assigned neighborhoods based off references and replaced their greybox layouts with the appropriate kits.

How did you generate the landscape to support your environment?

Any time you're dealing with large-scale terrain, World Machine will usually enter the equation. It's used either first to quickly create a realistic base mesh to refine and establish gameplay areas or to simply add more organic shapes and erosion to previously hand-sculpted terrain. Even in the latter case, there's usually plenty of manual hand-sculpting work within the editor.

In general, you want to avoid having large areas of one texture that tiles endlessly because using masks that are generated based off slope (or other parameters to blend multiple textures) will immediately assist in creating a realistic effect across the terrain.

As with neighborhoods that use specific building kits and recipes, having a strongly established biome for different areas of the world based upon elevation, location and climate can really help create a diverse palette and unique feel for every area of a game. Compare, for example, the barren, rocky desert of Silicon Valley with the redwood forests that surround San Francisco. Far Cry is another example of an open world that relies heavily upon distinct biomes to add variety to the art style of the game.

Here's another tip: instead of just relying upon tiled textures to add detail to your distant terrain, create a large-scale, diffuse map from World Machine that features all of the erosion detail baked into a single map. At a certain distance, your terrain material will fade from your detailed, tiling textures into a diffuse map that contains only large-scale details. This helps eliminate visual noise and shows only the large shapes and forms in your terrain. By doing so, you will help players read things much easier from a distance.

Could you discuss the overall composition of the city?

Nowadays, open worlds are massive. If we were hand placing every single bit of grass and decal in the world, it would take an endless amount of time. But, because we have tech artists and programmers, we're capable of minimizing the amount of work that's required to do everything on such a massive scale.

For the roads, we use splines that are drawn onto the terrain, where we can specify how wide they are or what material they're using based off pre-existing recipes. From there we can add additional, complex recipes to each road spline to specify what types of props that we would like to procedurally spawn on them, which randomizes each one or controls where they begin. Using this system, we can spawn garbage cans that are placed at every intersection, decals that randomly cover every road and street lights that are spaced apart every 15 meters.

In terms of architecture, the content team will generally do all of the modeling and texturing of the modular building pieces. After they're completed, they'll get integrated into the editor's building tool. Using this tool, we can create top-down shapes for our buildings that spawn different building-kit recipes on each edge of our shape. By adding more detail and cuts to the shape of the building, we can create endless building variations. These detailed buildings are then used to replace the 3D grey blocks that we used on our 2D maps created during pre-production. This is another reason why having strong foundational planning is so critical to the overall success of a project.

With the endless scope of these games, it’s important that all of these systems work well together and are as user friendly as possible, allowing quicker iteration and less manual, repetitive work. For example, when changing a road spline, all of the props that procedurally spawn on it will move along with it, alleviating the need to manually tweak thousands of meshes by hand.

The final critical element is strong, interdepartmental communication. Designers, tech artists, programmers and level artists absolutely need to communicate efficiently between each other to get the most out of every tool and system at their disposal. The more cohesive and free flowing communication is the stronger the final product will be.